Human nutrition

30 questions: 15 multiple choice, 15 structured.

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1 · Part 1 Core Diet [1]

Which row correctly shows a nutrient, a good dietary source of it and its importance in the diet?

	nutrient	dietary source	importance
A	iron	milk	strong bones and teeth
B	vitamin D	oily fish	helps the body absorb calcium for bones and teeth
C	vitamin C	citrus fruits	helps the body absorb calcium for bones and teeth
D	fibre	lean meat	prevents constipation

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2 · Part 1 Core Diet [1]

The bones of a young child become soft and weak, and the leg bones start to bend under the child's body weight.

Which dietary cause could lead to this condition?

A lack of vitamin C

B lack of iron

C too much fat

D lack of vitamin D

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3 · Part 1 Core Digestive system [1]

Which organ stores bile before the bile is released into the duodenum?

A gall bladder

B liver

C pancreas

D salivary gland

4 · Part 1 Core Physical digestion [1]

Which statement describes physical digestion?

A the breakdown of large insoluble molecules into small soluble molecules

B the use of enzymes to break down protein into amino acids

C the breakdown of food into smaller pieces without chemical change to the food molecules

D the taking of food and drink into the body through the mouth

5 · Part 1 Core Physical digestion [1]

Which row correctly matches a type of human tooth with its main function in physical digestion?

	type of tooth	main function
A	incisor	cutting and biting off pieces of food
B	canine	grinding and crushing food
C	molar	cutting and biting off pieces of food
D	premolar	biting off pieces of food

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6 · Part 1 Core Physical digestion [1]

Which list gives the tissues of the crown of a human tooth in order, starting from the outside and moving towards the centre?

A dentine, enamel, pulp

B enamel, dentine, pulp

C enamel, pulp, dentine

D pulp, dentine, enamel

7 · Part 1 Core Physical digestion [1]

How does the stomach carry out physical digestion?

A It secretes protease, which breaks down protein.

B It secretes hydrochloric acid, which kills harmful microorganisms.

C It absorbs most of the water from the food.

D Muscles in its wall contract to churn the food and mix it with gastric juice.

8 · Part 1 Extended Physical digestion [1]

What is the role of bile in the digestion of fats?

A It is an enzyme that breaks down fats into fatty acids and glycerol.

B It breaks fat molecules into fatty acids and glycerol without an enzyme.

C It breaks large fat droplets into smaller droplets, increasing the surface area for lipase.

D It carries fats from the small intestine into the lacteals.

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9 · Part 1 Core Chemical digestion [1]

Which row correctly shows an enzyme, the substance it breaks down and the product or products?

	enzyme	substance broken down	product or products
A	amylase	protein	amino acids
B	lipase	fats and oils	fatty acids and glycerol
C	protease	starch	simple reducing sugars
D	amylase	starch	amino acids

10 · Part 1 Core Chemical digestion [1]

Which row shows where lipase is secreted and where it acts?

	where lipase is secreted	where lipase acts
A	pancreas	small intestine
B	small intestine	stomach
C	liver	small intestine
D	pancreas	stomach

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11 · Part 1 Core Chemical digestion [1]

If the stomach of a person stopped producing hydrochloric acid, which change would happen?

A Fat droplets would no longer be emulsified in the stomach.

B Protein would be digested faster in the stomach.

C Less water would be absorbed in the stomach.

D More harmful microorganisms in food would survive.

12 · Part 1 Extended Chemical digestion [1]

Starch is digested to glucose in two stages.

Which row names the enzyme for each stage?

	enzyme that breaks down starch to maltose	enzyme that breaks down maltose to glucose
A	maltase	amylase
B	amylase	protease
C	amylase	maltase
D	lipase	maltase

13 · Part 1 Extended Chemical digestion [1]

Four test-tubes were set up at 37 °C. Each tube contained the same volume of a cloudy suspension of egg-white protein, the same volume of a protease solution and a solution to keep the pH constant. The suspension goes clear as the protein is digested.

tube	protease	pH
1	pepsin	2
2	pepsin	8
3	trypsin	2
4	trypsin	8

In which two tubes would the suspension go clear fastest?

A 1 and 4

B 1 and 3

C 2 and 3

D 2 and 4

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14 · Part 1 Core Absorption [1]

In which region of the alimentary canal are nutrients such as glucose and amino acids absorbed into the blood?

A stomach

B small intestine

C colon

D oesophagus

15 · Part 1 Extended Absorption [1]

Which feature of the epithelial cells covering a villus increases the internal surface area of the small intestine?

A many mitochondria

B a large nucleus

C a partially permeable cell membrane

D microvilli

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16 · Part 1 Core Diet [5]

Leon is 15 years old. For lunch on most days he eats white bread with butter and cheese, and he drinks a sugary fizzy drink.

(a) Describe what is meant by a balanced diet. [2]

(b) Leon's lunch contains very little fibre.

(i) State one good dietary source of fibre. [1]

(ii) Describe why fibre is important in the diet. [1]

(c) State one reason why the body needs water. [1]

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17 · Part 1 Core Diet [6]

Scientists at a remote research station eat mainly tinned meat, rice, pasta and white bread for several months. Fresh fruit and vegetables are not available.

(a) State the name of the deficiency disease the scientists are most likely to develop, and state its cause. [2]

(b) When a supply ship arrives, potatoes and oranges become available. The table shows the vitamin C content of these two foods.

food	vitamin C content / mg per 100 g
potato	20
orange	50

The recommended intake of vitamin C for an adult is 40 mg per day.

One scientist eats 120 g of potato in a day. Calculate the mass of orange this scientist must also eat that day to take in 40 mg of vitamin C in total. Show your working. [3]

(c) State one good dietary source of vitamin D. [1]

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18 · Part 1 Core Digestive system [6]

The digestive system carries out five processes: ingestion, digestion, absorption, assimilation and egestion.

(a) Name the process described by each statement.

(i) Muscle cells use amino acids to make new proteins. [1]

(ii) Undigested food leaves the body through the anus as faeces. [1]

(iii) Glucose moves from the small intestine into the blood. [1]

(b) State the function of the oesophagus. [1]

(c) Fibre in food is not digested in the human alimentary canal. Explain why fibre is egested instead of being absorbed. [2]

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19 · Part 1 Core Physical digestion [6]

During a dental check-up, a dentist examines a patient's teeth. Each tooth is embedded in the jaw bone and the gums.

(a) Name the part of a tooth that:

(i) contains nerves and blood vessels [1]

(ii) covers the root and helps to fix the tooth in the jaw bone. [1]

(b) Describe the function in physical digestion of:

(i) canines [1]

(ii) premolars. [1]

(c) Explain how chewing food helps chemical digestion. [2]

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20 · Part 1 Extended Physical digestion [6]

Milk contains fat. Two test-tubes, P and Q, were set up at 35 °C. A few drops of dilute sodium carbonate solution were added to each tube so that the starting pH was 8.0. A pH probe in each tube recorded the time taken for the pH to fall from 8.0 to 7.0.

tube	contents	time for pH to fall from 8.0 to 7.0 / min
P	10 cm^3 milk, 5 cm^3 lipase solution, 2 cm^3 distilled water	15
Q	10 cm^3 milk, 5 cm^3 lipase solution, 2 cm^3 bile salts solution	6

Bile salts are the substances in bile that emulsify fats. They do not contain any enzymes.

(a) Name the two products formed when lipase digests fat. [1]

(b) Suggest why the pH in each tube fell. [1]

(c) Calculate the percentage decrease in the time taken for the pH to fall when bile salts were added. [2]

(d) Outline how the bile salts caused the difference in time between tube P and tube Q. [2]

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21 · Part 1 Core Chemical digestion [4]

Food contains large molecules such as starch and protein. These must be chemically digested.

(a) Describe what is meant by chemical digestion. [2]

(b) State why the products of chemical digestion need to be small and soluble. [1]

(c) Name the products of the chemical digestion of protein. [1]

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22 · Part 1 Core Chemical digestion [7]

The table gives information about two digestive enzymes. Some entries are missing and have been replaced by gap numbers 1 to 5.

enzyme	where it is secreted	where it acts	substance broken down	product
amylase	gap 1 and pancreas	gap 2 and small intestine	starch	gap 3
protease	gap 4 and pancreas	stomach and small intestine	gap 5	amino acids

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(a) Give the missing entries for gap 1, gap 2 and gap 4. [3]

(b) Give the missing entries for gap 3 and gap 5. [2]

(c) Amylase from the mouth is swallowed with food and reaches the stomach. Explain why this amylase stops working in the stomach. [2]

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23 · Part 1 Core Chemical digestion [5]

The pH of the contents of a person's stomach was measured just before a meal and at intervals after it. The meal was eaten immediately after the reading at 0 minutes.

time after meal / min	pH of stomach contents
0	2.0
15	5.0
30	4.0
60	3.0
90	2.5
120	2.0

(a) Name the acid in gastric juice that makes the stomach contents acidic. [1]

(b) Describe the changes in the pH of the stomach contents shown in the table. [2]

(c) Suggest why the digestion of protein in the stomach is slow 15 minutes after the meal. [2]

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24 · Part 1 Extended Chemical digestion [7]

Bread and rice contain a lot of starch.

(a) Name the product formed when amylase breaks down starch. [1]

(b) Name the enzyme that breaks down this product into glucose, and state precisely where in the alimentary canal this enzyme acts. [2]

(c) Some people produce less of the enzyme in (b) than normal. Suggest and explain the effect on the amount of glucose absorbed into their blood after a meal of rice. [2]

(d) Explain why starch must be digested before the body can absorb it. [2]

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25 · Part 1 Extended Chemical digestion [7]

Two protease enzymes, P and Q, were taken from different regions of the human alimentary canal. The rate at which each enzyme digested protein was measured at different pH values.

pH	rate of protein digestion by enzyme P / arbitrary units	rate of protein digestion by enzyme Q / arbitrary units
1	6	0
2	10	0
3	7	1
4	3	2
5	1	4
6	0	6
7	0	9
8	0	10
9	0	8

(a) One of the enzymes is pepsin. Identify which enzyme is pepsin and give a reason for your answer using the data. [2]

(b) The mixture of food and gastric juice leaving the stomach is acidic. Explain how bile allows enzyme Q to work in the duodenum. [3]

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(c) In one person, the tube that carries bile to the duodenum becomes blocked. Use the data to suggest and explain the effect on protein digestion in this person's duodenum. [2]

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26 · Part 1 Core Absorption [5]

Each day, 9.0 dm^3 of water enters the alimentary canal of an adult from food, drinks and digestive juices. The table shows how much of this water is absorbed in two regions of the alimentary canal. All the remaining water leaves the body in the faeces.

region of the alimentary canal	volume of water absorbed per day / dm^3
region R	7.2
colon	1.6

(a) Region R is also where nutrients are absorbed. Identify region R. [1]

(b) Calculate the percentage of the water entering the alimentary canal each day that is absorbed in region R. [2]

(c) Calculate the volume of water that leaves the body in the faeces each day. [1]

(d) During an illness, the colon absorbs much less water than normal. Suggest how this changes the faeces. [1]

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27 · Part 1 Extended Absorption [7]

The inner lining of the small intestine is covered with villi. The epithelial cells covering each villus have microvilli.

(a) Explain the significance of villi and microvilli for the small intestine. [3]

(b) In a model of the small intestine, a short section of lining has a surface area of 3.0 cm^2 if it is treated as completely flat. The villi increase this surface area by a factor of 8. The microvilli then increase the surface area by a further factor of 20.

Calculate the total surface area of this section of lining. [2]

(c) The epithelial cells of the villi contain many mitochondria. Suggest why. [2]

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28 · Part 1 Extended Absorption [5]

Villi line the inner wall of the ileum.

(a) Describe the structure of a villus. [4]

(b) Suggest why it is an advantage that the epithelium covering a villus is only one cell thick. [1]

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29 · Part 1 Extended Absorption [5]

Each villus contains a lacteal and a network of blood capillaries.

(a) State which of these two vessels absorbs:

(i) glucose and amino acids [1]

(ii) fatty acids and glycerol. [1]

(b) Blood flows continuously through the capillaries of a villus. Explain how this helps the absorption of glucose. [2]

(c) Suggest one effect on a person if the lacteals in their villi became blocked. [1]

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30 · Part 1 Extended Absorption [7]

The small intestine is made up of the duodenum and the ileum. The table gives information about three regions of the alimentary canal in one adult.

region	length / m	villi present
duodenum	0.30	yes
ileum	5.70	yes
colon	1.50	no

(a) Calculate the percentage of the total length of the small intestine that is made up by the ileum. [2]

(b) Use the table and your own knowledge to explain why most nutrients are absorbed in the ileum and not in the colon. [3]

(c) Some people have part of their ileum removed by surgery. Suggest two effects this may have on the person. [2]

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